

CSA0904 – PROGRAMMING FOR JAVA

NAME: Dharshan A N

REG NO: 192525370

C03 – AT3

1. Problem Statement:

Develop a Java program to validate the age of an employee before registration. If the entered age is less than 18 or greater than 60 generate an appropriate user-defined exception.

Requirements:

Read employee name and age.

Create InvalidAgeException.

Use throw and throws.

Handle the exception using try-catch.

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1094. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "Dharshan A N 192525370RA".

On the left sidebar, there are four questions listed:

- Q1 Employee Age Validator (User-Defined Exception) 25 marks
- Q2 Printer Queue (Inter-Thread Communication) 25 marks
- Q3 Multithreaded Number Processing 25 marks
- Q4 Hospital Patient Registration (Multiple Exception Handling) 25 marks

The main content area shows the "Problem Statement" and "Requirements" for the selected question (Q1).

Problem Statement:
Develop a Java program to validate the age of an employee before registration. If the entered age is less than 18 or greater than 60 generate an appropriate user-defined exception.

Requirements:
Read employee name and age.
Create InvalidAgeException.
Use throw and throws.
Handle the exception using try-catch.

The "Your Code" section shows the following Java code:

```
1 import java.util.Scanner;
2
3 class Main {
4
5     static class InvalidAgeException extends Exception {
6         InvalidAgeException(String msg) {
7             super(msg);
8         }
9     }
10
11     static void checkAge(int age) throws InvalidAgeException {
12         if (age < 18 || age > 60)
13             throw new InvalidAgeException("Age is not valid");
14     }
15 }
```

The "Input" section shows the user has entered "dharshan" for the name and "18" for the age.

The "Output" section shows the program's execution results:

```
Enter Employee Name: Enter Age:
Employee Registered
Name: dharshan
Age: 18
```

The bottom status bar shows the system temperature as 90°F, weather as "Partly cloudy", and the time as 19:44 on 13-08-2026.

2. Problem Statement:

Develop a Java program to simulate a shared printer used by multiple users. A user thread submits a document while a printer thread processes it.

Requirements:

Create producer and consumer threads.

Use a shared document queue/buffer.

Implement wait() and notify().

Ensure proper communication between the threads.

Display submission and printing status.

The screenshot displays the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS ENGINEERING logo, the course title "SIMATS Online Programming Lab", and the user's name "Dharshan A N" with a dropdown menu. The main content area is divided into several sections:

- Problem Statement:** Develop a Java program to simulate a shared printer used by multiple users. A user thread submits a document while a printer thread processes it.
- Requirements:**
 - Create producer and consumer threads.
 - Use a shared document queue/buffer.
 - Implement wait() and notify().
 - Ensure proper communication between the threads.
 - Display submission and printing status.
- Your Code:** A text editor showing the following Java code:

```
1 class Main {
2
3     static String doc = null;
4
5     static synchronized void submit(String d) throws InterruptedException {
6         while (doc != null)
7             Main.class.wait();
8
9         doc = d;
10        System.out.println("Submitted: " + doc);
11
12        Main.class.notify();
13    }
14 }
```
- Input:** A text input field containing "stdin input".
- Output:** A text area showing the program's output:

```
Submitted: Document 1
Printing: Document 1
Submitted: Document 2
Printing: Document 2
Submitted: Document 3
Printing: Document 3
```

On the left side, there is a sidebar with a list of questions and their marks. The first question is "Employee Age Validator" (25 marks). The second question is "Q2 Printer Queue" (25 marks), which is marked as completed. The third question is "Q3 Multithreaded Number Processing" (25 marks), also marked as completed. The fourth question is "Q4 Hospital Patient Registration" (25 marks). At the bottom of the sidebar, it says "4/4 answered".

3. Problem Statement:

Develop a Java program that creates multiple threads to perform different operations on a set of numbers.

Requirements:

Create separate threads to calculate the sum average and maximum value.

Use the Thread class.

Execute the threads concurrently.

Display the result produced by each thread

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt/assignment_id=1094`. The page header includes the SIMATS Engineering logo, the title "SIMATS Online Programming Lab", and the subject "JAVA PROGRAMMING". A user profile for "Dharshan A N" is visible in the top right corner.

Problem Statement:
Develop a Java program that creates multiple threads to perform different operations on a set of numbers.

Requirements:
Create separate threads to calculate the sum average and maximum value.
Use the Thread class.
Execute the threads concurrently.
Display the result produced by each thread

Your Code

```
1 class Main {
2
3     static int[] a = {10, 20, 30, 40, 50};
4
5     static class Sum extends Thread {
6         public void run() {
7             int s = 0;
8             for (int x : a)
9                 s += x;
10            System.out.println("Sum = " + s);
11        }
12    }
13
14    static class Average extends Thread {
15        public void run() {
```

Input
stdin input

Output
Maximum = 50
Average = 30
Sum = 150

The interface also features a sidebar on the left with a list of questions (Q1 to Q4) and their respective marks. Q1 is "Employee Age Validator" (25 marks), Q2 is "Printer Queue" (25 marks), Q3 is "Multithreaded Number Processing" (25 marks), and Q4 is "Hospital Patient Registration" (25 marks). A status "4/4 answered" is shown at the bottom of the sidebar.

4. Problem Statement:

Develop a Java program for hospital patient registration. The system should accept patient age number of visits and patient ID. Handle invalid input and invalid calculations using appropriate exception handling mechanisms.

Requirements:

Accept patient details.

Handle `NumberFormatException` and `ArithmeticException`.

Use multiple catch blocks.

Display meaningful error messages.

Use a finally block to indicate completion of registration.

The screenshot displays the SIMATS Online Programming Lab interface. The browser address bar shows the URL: `simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1094`. The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "Dharshan A N 192525370RA".

Problem Statement:
Develop a Java program for hospital patient registration. The system should accept patient age number of visits and patient ID. Handle invalid input and invalid calculations using appropriate exception handling mechanisms.

Requirements:
Accept patient details.
Handle `NumberFormatException` and `ArithmeticException`.
Use multiple catch blocks.
Display meaningful error messages.
Use a finally block to indicate completion of registration.

Your Code:

```
1 import java.util.*;  
2  
3 class Main {  
4     public static void main(String[] args) {  
5  
6         Scanner sc = new Scanner(System.in);  
7  
8         try {  
9             System.out.print("Enter Patient ID: ");  
10            String id = sc.nextLine();  
11  
12            System.out.print("Enter Age: ");  
13            int age = Integer.parseInt(sc.nextLine());
```

Input:
101
18
5

Output:
Enter Patient ID: Enter Age:
ID: 101
Age: 18
Visits: 5
Calculation: 3

The interface also shows a sidebar with a list of questions (Q1 to Q4) and a status bar at the bottom indicating the temperature (90°F) and time (19:45, 13-08-2026).